

Assignment

CSA5113 -

Cryptography

Karthheeka Repalle

192372289

CSE-A1

~~Set 3~~

1.

Triple DES (3DES) Encryption Using Two Keys.

Given

Key 1 = 101010

Key 2 = 010101

Key 3 = Data block = 110011

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Triple DES applies the DES operation three times to the plaintext.

when using two keys, the standard form is:

$$C = EK_1(DK_2(EK_1(P)))$$

Encrypt $K_1 \rightarrow$ Decrypt $K_2 \rightarrow$ Encrypt K_1

DES Encryption :- $D = 110011$

$$X = EK_1(D)$$

$$X = E_{101010}(110011)$$

DES Decryption :-

$$Y = DK_2(X)$$

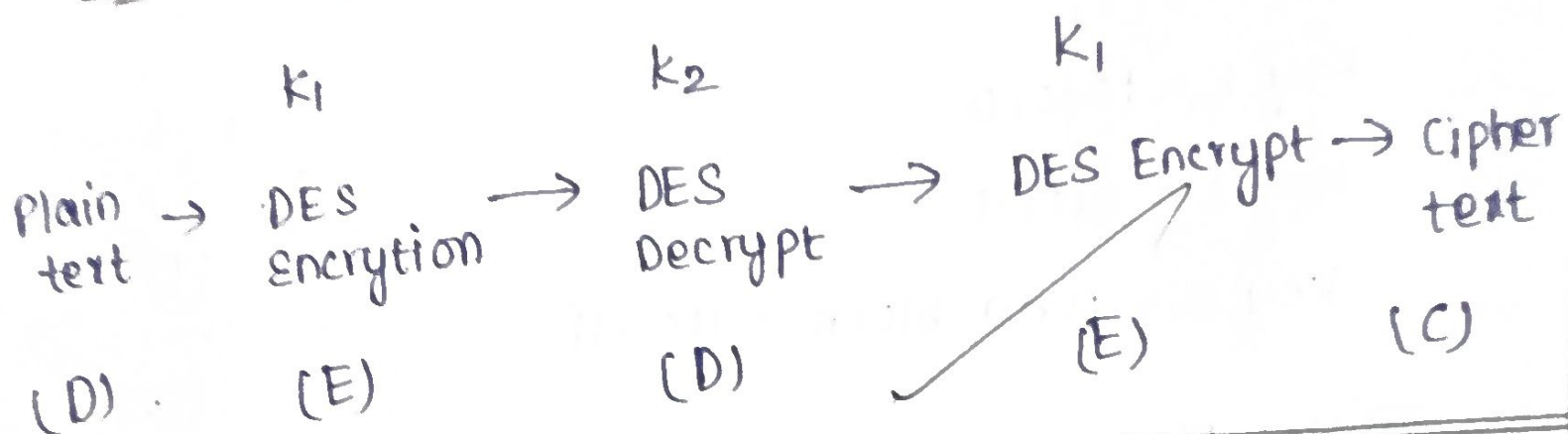
$$Y = D_{010101}(E_{101010}(110011))$$

DES encryption using K_1 :

$$C = E_{K_1}(Y)$$

$$C = E_{101010}(D_{010101}(E_{101010}(110011)))$$

3DES Daigram



2.

Meet in the middle Attack on Double DES

Double DES attempts to improve security by encryption the plaintext twice using two independent keys

$$C = E_{K_2}(E_{K_1}(P))$$

DES has 56-bit key

$$2^{56} \times 2^{56} = 2^{112} \text{ — key combinations}$$

meet in the middle is a cryptanalytic technique where the attacker works from both ends.

Double DES gives:

$$C = E_{K_2}(E_{K_1}(P))$$

$$\Rightarrow X = E_{K_1}(P)$$

$$C = E_{K_2}(X)$$

Public key contain (e, n)

$$\boxed{\text{Public key} = (3, 33)}$$

(ii) calculate Euler's phi

$$\phi(n) = (p-1)(q-1)$$

$$\phi(33) = (3-1)(11-1)$$

$$= 2 \times 10$$

$$= 20$$

(iii) Private Exponent (d)

$$ed = 1 \pmod{\phi(n)}$$

$$3d = 1 \pmod{20}$$

$$\text{Try } d=7 \Rightarrow 3 \times 7 = 21$$

$$21 \pmod{20} = 1$$

$$\boxed{d=7}$$

$$\text{Private key } \boxed{(d, n) = (7, 33)}$$

RSA Encryption

$$C = M^e \pmod{n}$$

$$\Rightarrow C = 5^3 \pmod{33}$$

$$\rightarrow 5^3 = 125$$

$$125 \pmod{33} = 26$$

$$125 = 3(33) + 26$$

$$\boxed{C=26}$$

$$M=5$$

$$e=3$$

$$n=33$$

encrypted message is 26.

$$x = DK_2(c)$$

$$EK_1(P) = DK_2(c)$$

Procedure:-

$$x_i = EK_1(P)$$

$$\text{Decrypt} \Rightarrow y_j = DK_2(c)$$

$$\text{Find a match} \rightarrow EK_1(P) = DK_2(c) = x$$

$$\Rightarrow (K_1, K_2)$$

Forward direction

$$x \leftarrow E(K_1) \leftarrow P$$

↑
match

$$x \xrightarrow{E(K_2)} c$$

Backward direction

3. RSA Encryption and Decryption

Given

$$p = 3$$

$$q = 11$$

$$e = 3$$

$$M = 5$$

(i) RSA calculates

$$n = pq$$

$$n = 3 \times 11$$

$$n = 33$$

RSA Decryption

$$M = c^d \bmod n$$

$$M = 26^7 \bmod 33$$

$$\Rightarrow 26^2 = 676$$

$$\Rightarrow 676 \bmod 33 = 16$$

$$26^4 \bmod 33 = 16^2 \bmod 33$$

$$= 256 \bmod 33 = 25$$

$$\Rightarrow 25 \times 16 = 400$$

$$400 \bmod 33 = 4$$

$$\text{Then: } 4 \times 26 = 104$$

$$104 \bmod 33 = 5$$

$$\boxed{M=5}$$

Decrypted message = 5

$$\Rightarrow 5 \rightarrow 26 \rightarrow 5$$

4.

Padding in Block Ciphers

Padding is the process of adding extra bytes/bits to plaintext so that its length becomes an exact multiple of cipher's block size.

AES $\rightarrow$ 16-byte blocks

DES $\rightarrow$ 3DES $\rightarrow$ 8-byte blocks.

HELLO - 5 bytes - block size 8 bytes

$8 - 5 = 3$ - additional bytes

SO, HELLO 08 03 03

In hexadecimal 48 45 4C 4C 4F 03 03 03

5. Elliptic Curve Cryptography (ECC) for secure message in IOT Devices.

elliptic curve is $y^2 = x^3 + ax + b$

256 ECC $\approx$ 3072-bit RSA

(i) key generation :- Private key - d_A, d_B

Public key - Q_A, Q_B

(ii) Generate shared Secret :- $S = d_A Q_B$

$$S = d_B Q_A$$

$$\Rightarrow d_A Q_B = d_B Q_A$$

(iii) Symmetric Encryption key :- $K = \text{KDF}(S)$

IOT Communication Architecture :-

ECC / ECDH

